

	be 1 m and not 2 m.
15.	Ans: C Separation between the horizontal threads is determined by the vertical bright fringes. $\tan \theta = \frac{0.0548 / 2}{2.00} = 0.0137$ $\theta = 0.7849^{\circ}$

$$d \sin \theta = m\lambda$$

$$d \sin 0.7849^\circ = 2(685 \times 10^{-9})$$

$$\mathbf{d = 1.00 \times 10^{-4} \text{ m} = 0.100 \text{ mm.}}$$

For comparison, the spacing of the vertical threads is determined by the horizontal bright fringes:

$$d \sin 0.7849^\circ = 1(685 \times 10^{-9})$$

$$d = 5.00 \times 10^{-5} \text{ m} = 0.0500 \text{ mm.}$$